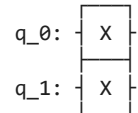


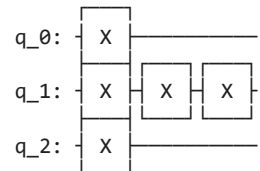
```
!pip install qiskit
```

```
Requirement already satisfied: qiskit in /usr/local/lib/python3.13/dist-packages (2.5.2)  
Requirement already satisfied: numpy<3,>=2.0 in /usr/local/lib/python3.13/dist-packages (from qiskit) (2.1.3)  
Requirement already satisfied: scipy>=1.14 in /usr/local/lib/python3.13/dist-packages (from qiskit) (1.16.3)  
Requirement already satisfied: rustworkx>=0.15.0 in /usr/local/lib/python3.13/dist-packages (from qiskit) (0.18.1)  
Requirement already satisfied: dill>=0.3 in /usr/local/lib/python3.13/dist-packages (from qiskit) (0.4.1)  
Requirement already satisfied: stevedore>=3.0.0 in /usr/local/lib/python3.13/dist-packages (from qiskit) (5.9.1)  
Requirement already satisfied: typing-extensions in /usr/local/lib/python3.13/dist-packages (from qiskit) (4.16.0)
```

```
from qiskit import QuantumCircuit as qc  
a=qc(2)  
a.x(0)  
a.x(1)  
print(a)
```



```
from qiskit import QuantumCircuit as qc  
a=qc(3)  
a.x(1)  
a.x(1)  
a.x(1)  
a.x(0)  
a.x(2)  
print(a)
```



```
from qiskit import QuantumCircuit as qc  
a=qc(1)  
a.y(0)  
print(a)  
a=qc(3)  
a.y(0)  
a.y(1)  
print(a)
```

q: 

q_0: 

q_1: 

q_2: 

```
from qiskit import QuantumCircuit as qc
a=qc(1)
a.x(0)
a.y(0)
print(a)
```

q: 

```
from qiskit import QuantumCircuit as qc
b=qc(3)
b.z(2)
print(b)
```

q_0: 

q_1: 

q_2: 

```
b=qc(3)
b.z(0)
b.z(1)
b.z(2)
print(b)
```

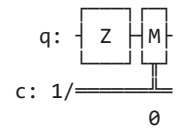
q_0: 

q_1: 

q_2: 

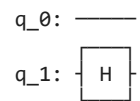
```
from qiskit import QuantumCircuit as qc
a=qc(1,1)
a.z(0)
```

```
a.measure(0,0)
print(a)
```



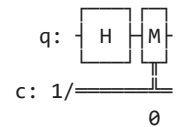
```
from qiskit import QuantumCircuit as qc
a=qc(2)
a.h(1)

print(a)
```

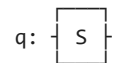


```
a=qc(1,1)
a.h(0)
a.measure(0,0)

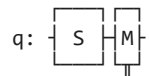
print(a)
```



```
from qiskit import QuantumCircuit as qc
a=qc(1)
a.s(0)
print(a)
```



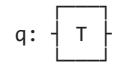
```
from qiskit import QuantumCircuit as qc
a=qc(1,1)
a.s(0)
a.measure(0,0)
print(a)
```



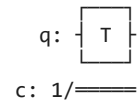
c: 1/══════

0

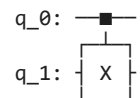
```
from qiskit import QuantumCircuit as qc
a=qc(1)
a.t(0)
print(a)
```



```
from qiskit import QuantumCircuit as qc
a=qc(1,1)
a.t(0)
print(a)
```

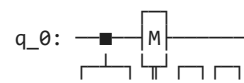
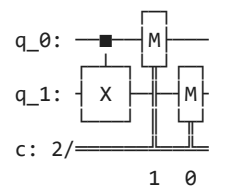


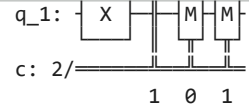
```
from qiskit import QuantumCircuit as qc
a=qc(2)
a.cx(0,1)
print(a)
```



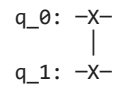
```
from qiskit import QuantumCircuit as qc
a=qc(2,2)
a.cx(0,1)
a.measure([0 ,1],[1 ,0])
print(a)
```

```
a.measure(1,1)
print(a)
```

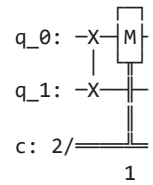




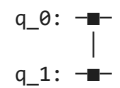
```
from qiskit import QuantumCircuit as qc
a=qc(2)
a.swap(1,0)
print(a)
```



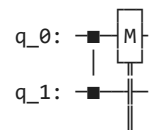
```
from qiskit import QuantumCircuit as qc
a=qc(2,2)
a.swap(0,1)
a.measure(0,1)
print(a)
```



```
from qiskit import QuantumCircuit as qc
a=qc(2)
a.cz(0,1)
print(a)
```

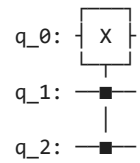


```
from qiskit import QuantumCircuit as qc
a=qc(2,2)
a.cz(0,1)
a.measure(0,1)
print(a)
```



c: 2/1

```
from qiskit import QuantumCircuit as qc
a=qc(3)
a.ccx(1,2,0)
print(a)
```



```
from qiskit import QuantumCircuit as qc
a=qc(3,3)
a.ccx(0,1,2)
a.measure([0,1,2],[0,1,2])
print(a)
```

